

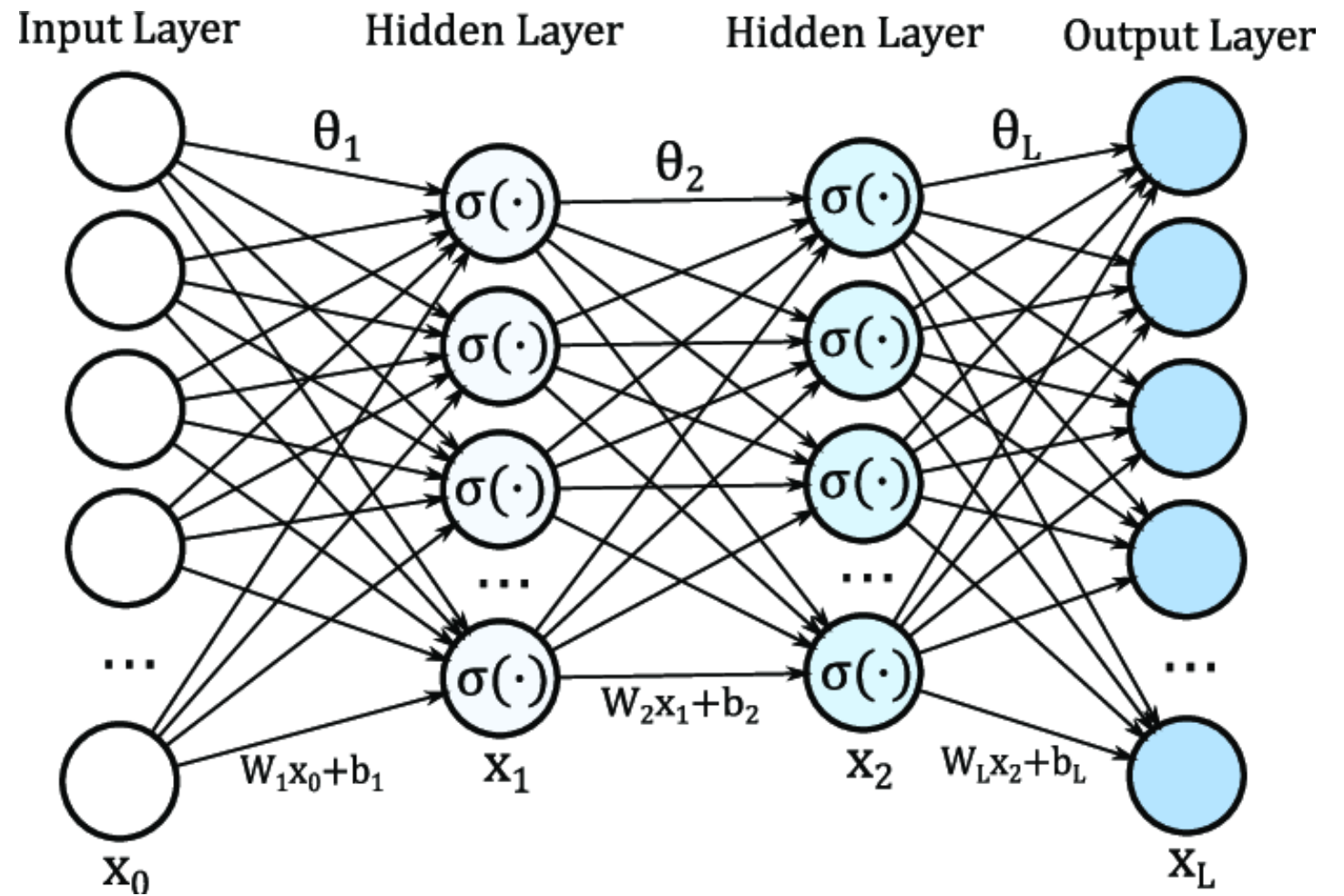
Deep Learning Text Processing

Deep Learning

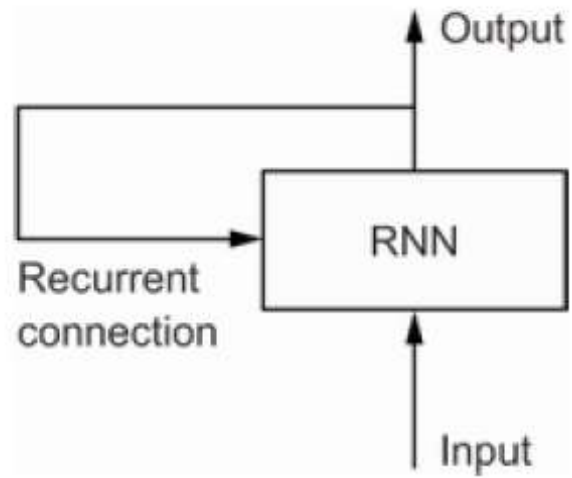
Sher Muhammad Daudpota

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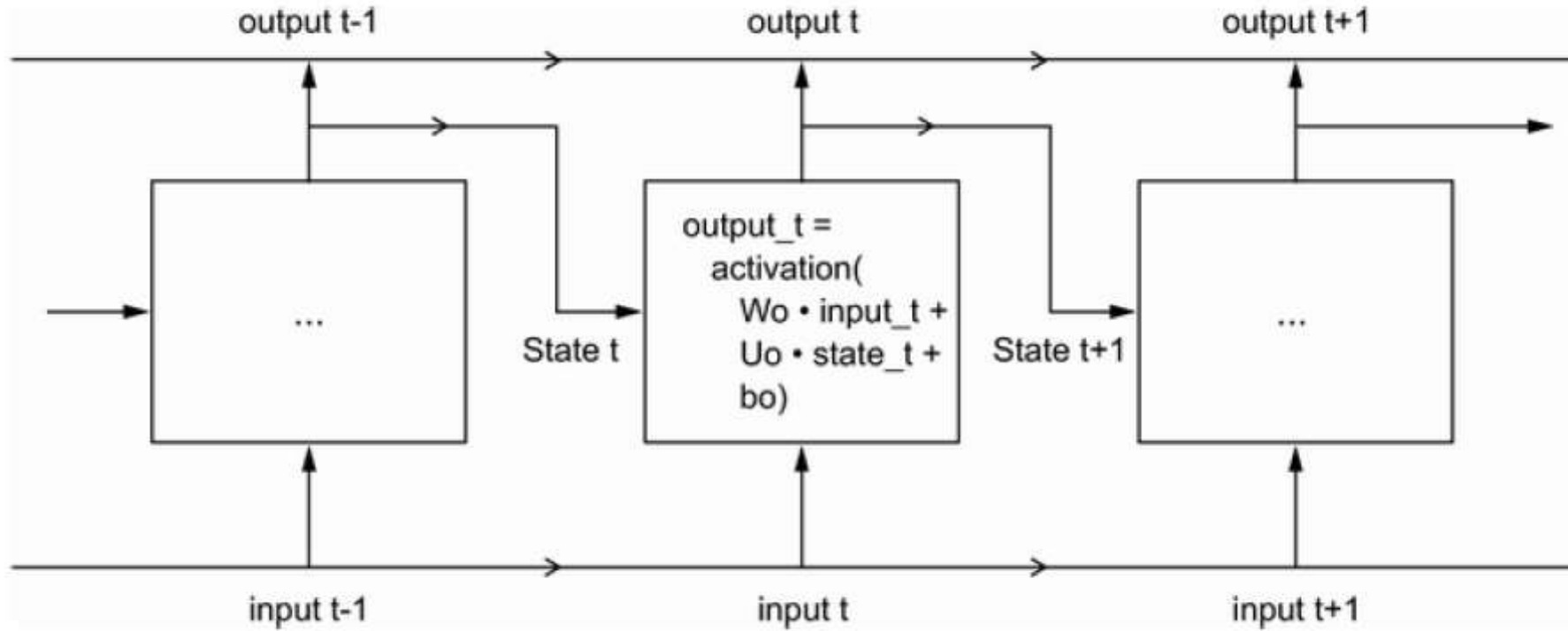
Revision of Fully Connected Neural Network



Recurrent Neural Network



Recurrent Neural Network



LSTM

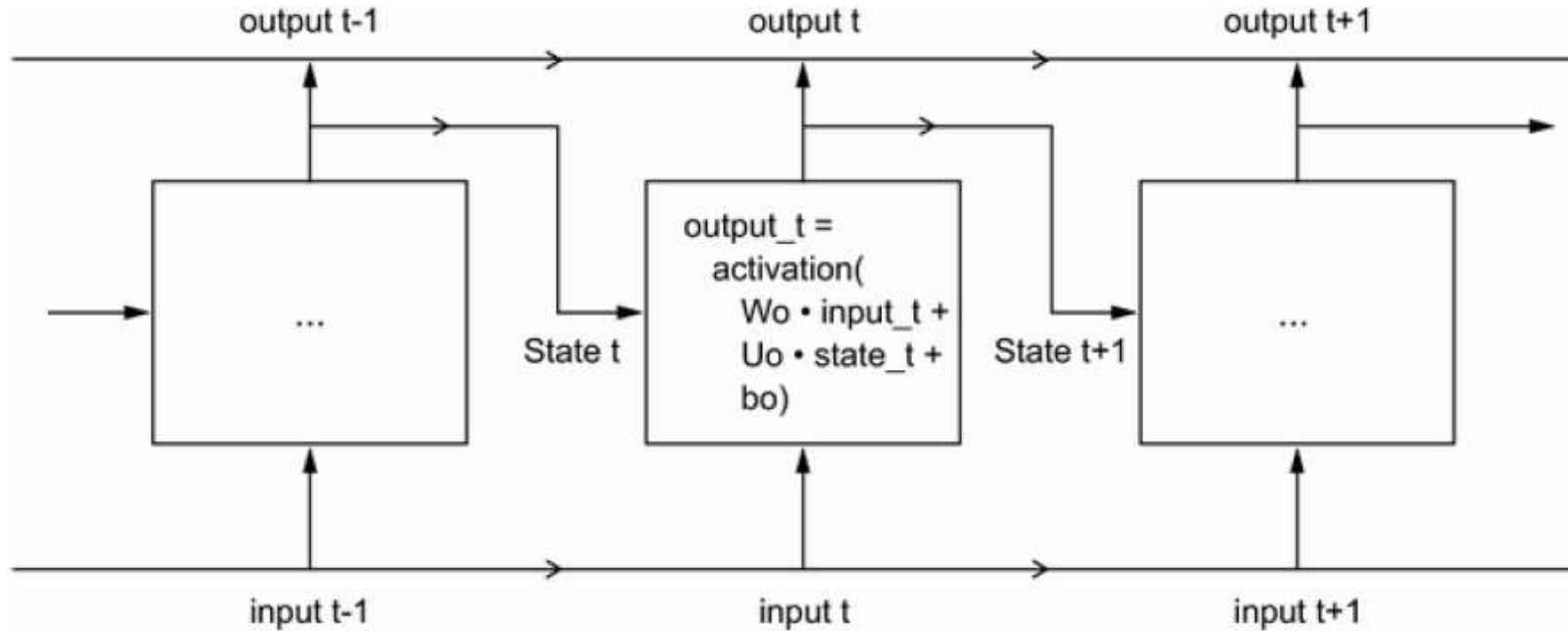


Figure 10.8 The starting point of an LSTM layer: a SimpleRNN

LSTM

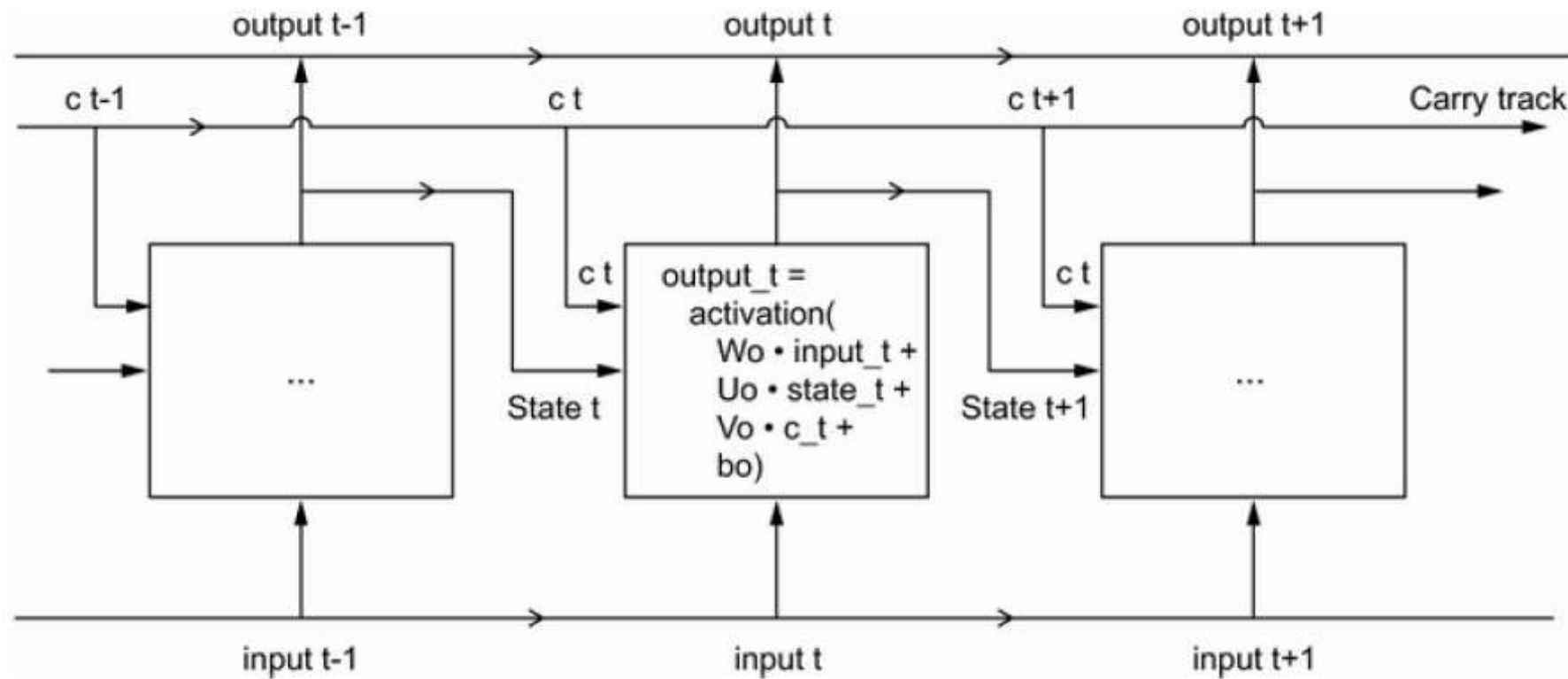


Figure 10.9 Going from a SimpleRNN to an LSTM: adding a carry track

LSTM

```
output_t = activation(dot(state_t, Uo) + dot(input_t, Wo) + dot(c_t, Vo) + bo)
i_t = activation(dot(state_t, Ui) + dot(input_t, Wi) + bi)
f_t = activation(dot(state_t, Uf) + dot(input_t, Wf) + bf)
k_t = activation(dot(state_t, Uk) + dot(input_t, Wk) + bk)
```

$$c_{t+1} = i_t * k_t + c_t * f_t$$

LSTM

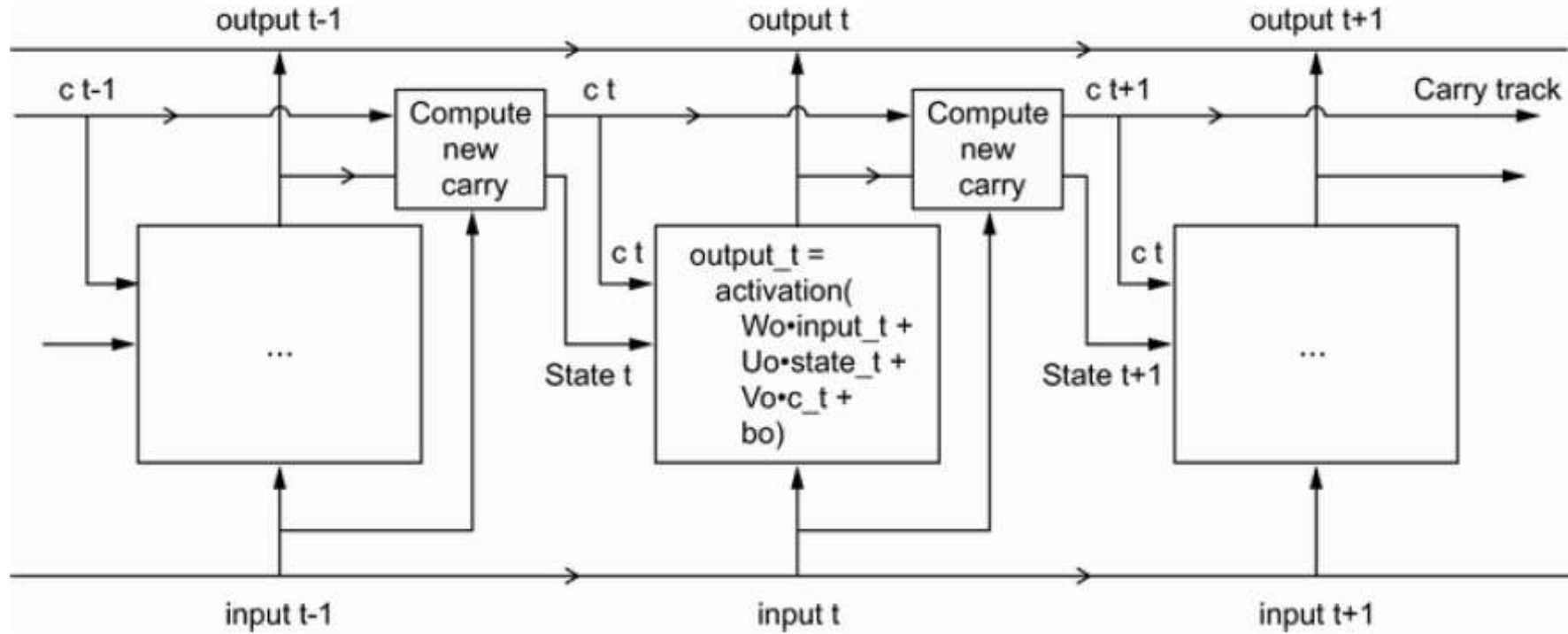


Figure 10.10 Anatomy of an LSTM

Bidirectional RNN

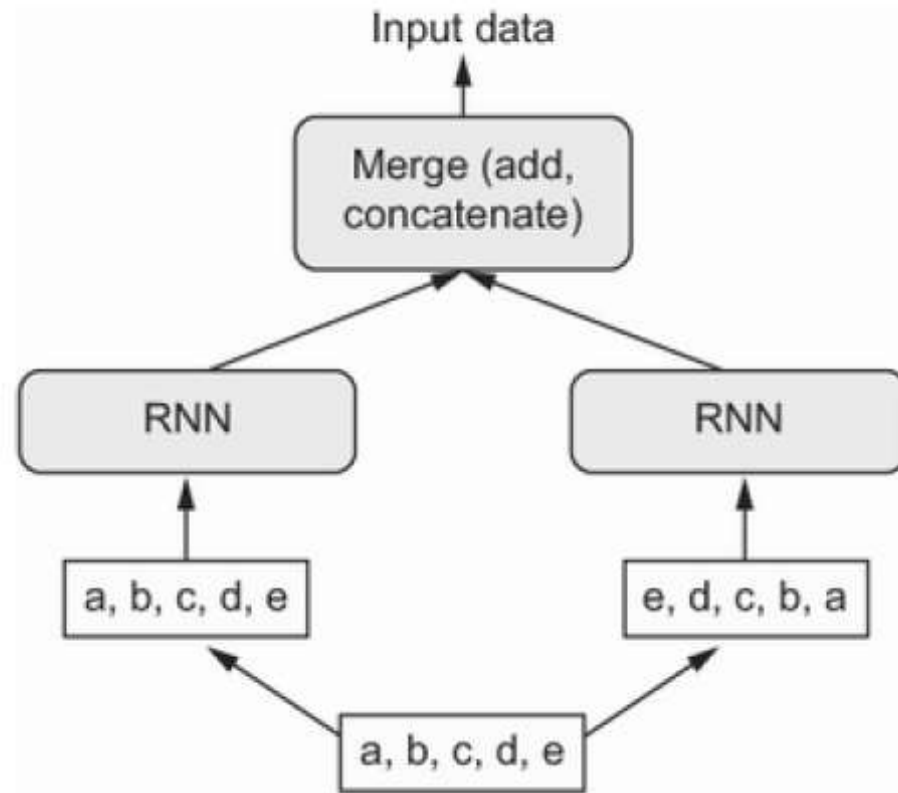


Figure 10.14 How a bidirectional RNN layer works